

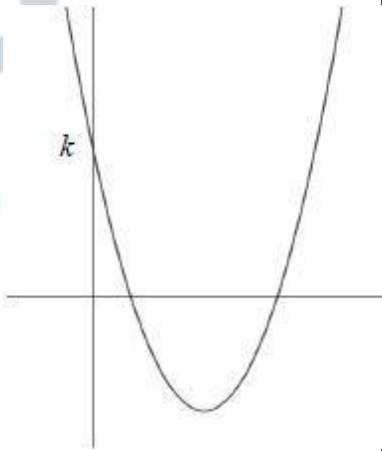
## Mark schemes

### Q1.

|     | Marking Instructions                                                                                                                                                                                            | AO     | Marks | Typical Solution                                                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | Takes out a factor of 2, or obtains $a = 2$ by equating coefficients.<br>Note $2(x^2 - \text{'anything'})$ or $2(x - \text{'anything'})^2$ scores M1                                                            | AO1.1a | M1    | $2\left(x^2 - \frac{5x}{2}\right) + k$ $2\left(x - \frac{5}{4}\right)^2 + k - \frac{25}{8}$                                                    |
|     | Expresses as $\left(x - \frac{5}{4}\right)^2$ or obtains $b = \frac{5}{4}$ by equating coefficients                                                                                                             | AO1.1a | M1    |                                                                                                                                                |
|     | Obtains correct expression. If using equating coefficients <b>must</b> be put back in given form required                                                                                                       | AO1.1b | A1    |                                                                                                                                                |
| (b) | Selects a method using completed square form (recognition that vertex occurs when $x = \text{'their' } b$ ), discriminant (any use of $b^2 - 4ac$ seen) or calculus (finds $y$ coordinate of stationary point). | AO3.1a | M1    | Typical solution 1<br>$\text{When } x = \frac{5}{4} \left(x - \frac{5}{4}\right)^2 = 0$ $\left(k - \frac{25}{8}\right) > 3$ $k > \frac{49}{8}$ |
|     | Forms an appropriate correct inequality for their chosen method. (first time inequality sign seen in each typical solution scores A1)                                                                           | AO1.1b | A1    |                                                                                                                                                |
|     | Obtains $k > \frac{49}{8}$ (ACF)(OE)                                                                                                                                                                            | AO1.1b | A1    |                                                                                                                                                |
|     |                                                                                                                                                                                                                 |        |       | Typical solution 2<br>$2x^2 - 5x + k = 3$ $2x^2 - 5x + k - 3 = 0$ $5^2 - 4 \times 2 \times (k - 3) < 0$                                        |

|                      |  |  |                                                                                                                                                                                                                                 |
|----------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |  |  | $k > \frac{49}{8}$                                                                                                                                                                                                              |
|                      |  |  | Typical solution 3<br>$\frac{dy}{dx} = 4x - 5$<br>At stationary point $\frac{dy}{dx} = 0$<br>$4x - 5 = 0$<br>$x = \frac{5}{4}$<br>when $x = \frac{5}{4}$ $y = k - \frac{25}{8}$<br>$k - \frac{25}{8} > 3$<br>$k > \frac{49}{8}$ |
| <b>Total 6 marks</b> |  |  |                                                                                                                                                                                                                                 |

**Q2.**

|     | Marking Instructions                                                                         | AO     | Marks | Typical Solution                                                                                        |
|-----|----------------------------------------------------------------------------------------------|--------|-------|---------------------------------------------------------------------------------------------------------|
| (a) | Sketches graph recalling correct u shape                                                     | AO1.2  | B1    |                    |
|     | Deduces correct relative positions of intersections with axes and with $k$ labelled          | AO2.2a | B1    |                                                                                                         |
| (b) | Shows evidence of discriminant being used or completing the square to find vertex            | AO1.1a | M1    | For distinct roots<br>$b^2 - 4ac > 0$<br>$(-6)^2 - 4 \times 1 \times k > 0$<br>$36 - 4k > 0$<br>$k < 9$ |
|     | Obtains $k < 9$<br>Condone $k \leq 9$                                                        | AO1.1b | A1    |                                                                                                         |
|     | Explains that positive roots and the u shape of the graph (OE) mean the graph must cross the | AO2.4  | E1    |                                                                                                         |

|                                        |        |    |             |
|----------------------------------------|--------|----|-------------|
| $y$ -axis above 0 or $k > 0$ .         |        |    | $0 < k < 9$ |
| States correct range of values for $k$ | AO2.2a | R1 |             |
| Total 6 marks                          |        |    |             |

**Q3.**

| Marking Instructions                                                                                                                                                                                         | AO     | Marks | Typical Solution                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selects an appropriate method – <b>either</b> differentiates, at least as far as:<br>$\frac{dy}{dx} = 2x \dots$<br><b>or</b> commences completion of the square:<br>$\left(x - \frac{5}{2}\right)^2 + \dots$ | AO1.1a | M1    | $y = \left(x - \frac{5}{2}\right)^2 - \frac{25}{4} + a$<br>$y$ minimised when squared bracket is 0<br>$\left(\frac{5}{2}, a - \frac{25}{4}\right)$<br><b>ALT</b><br>$\frac{dy}{dx} = 2x - 5$<br>so $2x - 5 = 0$ for minimum<br>$x = \frac{5}{2}$ |
| Fully differentiates <b>and</b> sets derivative equal to zero <b>or</b> fully completes square<br>Allow one error                                                                                            | AO1.1a | M1    | $y = \left(\frac{5}{2}\right)^2 - 5\left(\frac{5}{2}\right) + a = a - \frac{25}{4}$                                                                                                                                                              |
| Obtains both coordinates                                                                                                                                                                                     | AO1.1b | A1    |                                                                                                                                                                                                                                                  |
|                                                                                                                                                                                                              |        |       | <b>Total 3 marks</b>                                                                                                                                                                                                                             |

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**Q4.**

| Marking Instructions                                       | AO     | Marks | Typical Solution                                                              |
|------------------------------------------------------------|--------|-------|-------------------------------------------------------------------------------|
| Forms discriminant – condone one error in discriminant     | AO1.1a | M1    | for distinct real roots, disc $> 0$<br>$4^2 - 4 \times 3 \times (2k - 1) > 0$ |
| States that discriminant $> 0$ for real and distinct roots | AO2.4  | R1    | $16 - 12(2k - 1) > 0$<br>$28 - 24k > 0$                                       |
| Forms an inequality from 'their' discriminant              | AO1.1a | M1    | $k < \frac{7}{6}$                                                             |
| Solves inequality for $k$ correctly                        | AO1.1b | A1    |                                                                               |

|                                         |  |  |  |
|-----------------------------------------|--|--|--|
| Allow un-simplified equivalent fraction |  |  |  |
| <b>Total 4 marks</b>                    |  |  |  |

**Q5.**

| Marking Instructions                                             | AO     | Marks | Typical Solution                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------|--------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Divides or multiplies by $\cos \theta$                           | AO3.1a | M1    | $\frac{\sin \theta \tan \theta}{\cos \theta} + 2 \frac{\sin \theta}{\cos \theta} = 3$                                                                                                                                                                                                         |
| Obtains correct quadratic                                        | AO1.1b | A1    | $\tan^2 \theta + 2 \tan \theta - 3 = 0$                                                                                                                                                                                                                                                       |
| Applies a correct method to solve 'their' quadratic PI           | AO1.1a | M1    | $(\tan \theta + 3)(\tan \theta - 1) = 0$<br>$\tan \theta = 1$ or $-3$                                                                                                                                                                                                                         |
| Finds two correct values of $\tan \theta$ from 'their' quadratic | AO1.1b | A1F   | $\theta = 45^\circ$ or $108^\circ$<br><b>ALT</b>                                                                                                                                                                                                                                              |
| Obtains two correct answers CAO                                  | AO1.1b | A1    | $\sin \theta \tan \theta \cos \theta + 2 \sin \theta \cos \theta = 3 \cos^2 \theta$<br>$\sin^2 \theta + 2 \sin \theta \cos \theta - 3 \cos^2 \theta = 0$<br>$(\sin \theta + 3 \cos \theta)(\sin \theta - \cos \theta) = 0$<br>$\tan \theta = 1$ or $-3$<br>$\theta = 45^\circ$ or $108^\circ$ |
| <b>Total 5 marks</b>                                             |        |       |                                                                                                                                                                                                                                                                                               |

**Q6.**

| Marking Instructions                                                                                                                                                                                                                                  | AO    | Marks | Typical Solution                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Explains clearly that $f(x)$ is increasing $\Leftrightarrow f'(x) > 0$ (for all values of $x$ )<br>or<br>Explains $\Rightarrow f(x)$ is increasing<br>$f'(x) > 0$ for all values of $x$<br>This may appear at any appropriate point in their argument | AO2.4 | E1    | For all $x$ , $f'(x) > 0 \Rightarrow f(x)$ is an increasing function<br>$f(x) = x^3 - 3x^2 + 15x - 1$<br>$\Rightarrow f'(x) = 3x^2 - 6x + 15$<br>$\Rightarrow f'(x) = 3(x-1)^2 + 12$<br>$\therefore f'(x)$ has a minimum value of 12 (*)<br><br>therefore $f'(x) > 0$ for all values of $x$ |

|                                                                                                                                                                                                          |        |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Differentiates – at least two correct terms                                                                                                                                                              | AO1.1a | M1 | <b>OR</b><br>for $f'(x)$ , $b^2 - 4ac = -144$<br>$\therefore f'(x) \neq 0$ for any real $x$ ,<br>so $f'(x)$ is either always positive or always negative.<br>$f'(0) = 15$<br>therefore $f'(x) > 0$ for all values of $x$<br><b>OR</b><br>$f''(x) = 6x - 6$ , which = 0 when $x = 1$<br>so min $f'(x)$ is $f'(1) = 12$<br>therefore $f'(x) > 0$ for all values of $x$<br><br>Thus, since, $f'(x) > 0$ for all values of $x$ it is proven that $f(x)$ is an increasing function. |
| All terms correct                                                                                                                                                                                        | AO1.1b | A1 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Attempts a correct method which could lead to $f'(x) > 0$                                                                                                                                                | AO3.1a | M1 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Correctly deduces $f'(x) > 0$<br>(for all values of $x$ )                                                                                                                                                | AO2.2a | A1 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Writes a clear statement that links the steps in the argument together, the deduction about a positive gradient for all values of $x$ proves that the given function is increasing for all values of $x$ | AO2.1  | R1 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                                                                                                                                                          |        |    | <b>Total 6 marks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

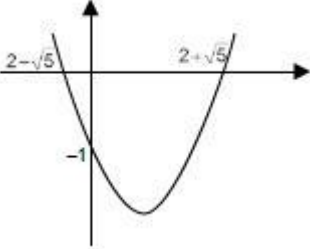
**Q7.**

| Marking Instructions                                                         | AO     | Marks | Typical Solution                                                                                                                |
|------------------------------------------------------------------------------|--------|-------|---------------------------------------------------------------------------------------------------------------------------------|
| Clearly states that equal roots $\Rightarrow b^2 - 4ac = 0$                  | AO2.4  | B1    | $\therefore b^2 - 4ac = 0$ for equal roots<br>$k^2 - 4(2k - 3)(k - 1) = 0$<br>$7k^2 - 20k + 12 = 0$<br>$k = \frac{6}{7}, k = 2$ |
| Forms quadratic expression in $k$<br>(allow one error)                       | AO1.1a | M1    |                                                                                                                                 |
| Obtains correct quadratic equation in $k$<br>(PI by correct values for $k$ ) | AO1.1b | A1    |                                                                                                                                 |
| Obtains correct values for $k$ for 'their' quadratic equation                | AO1.1b | A1F   |                                                                                                                                 |
|                                                                              |        |       | <b>Total 4 marks</b>                                                                                                            |

**Q8.**

|     | Marking Instructions                                                                                                                                                                                                                                                                      | AO     | Marks | Typical Solution                                                                                             |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|--------------------------------------------------------------------------------------------------------------|
| (a) | Completes the square twice or applies standard formula                                                                                                                                                                                                                                    | AO1.1a | M1    | $(x + 4)^2 + (y - 6)^2 - 16 - 36 = 12$<br>$(x + 4)^2 + (y - 6)^2 = 64$                                       |
|     | Obtains correct equation                                                                                                                                                                                                                                                                  | AO1.1b | A1    | Centre $(-4, 6)$                                                                                             |
|     | Obtains correct radius and correct coordinates of C<br>Follow through 'their' equation                                                                                                                                                                                                    | AO1.1b | A1F   | Radius = 8                                                                                                   |
| (b) | Demonstrates a method to find the length $OP$ or $OQ$ (or their squares), or the coordinates $P$ or $Q$ using 'their' values from part (a)                                                                                                                                                | AO3.1a | M1    | $OC^2 = 4^2 + 6^2 = 52$<br>$OP^2 = r^2 - OC^2$<br>$= 64 - 52 = 12$<br>$PQ = 2OP$<br>$2\sqrt{12} = 4\sqrt{3}$ |
|     | Uses a circle property that may lead to a solution, eg radius and chord meet at right-angles (evidence for this could be the use of Pythagoras or perpendicular gradients)                                                                                                                | AO3.1a | M1    |                                                                                                              |
|     | Finds $OP$ or $OQ$ or coordinates of $P$ or $Q$ <b>CAO</b>                                                                                                                                                                                                                                | AO1.1b | A1    |                                                                                                              |
|     | Obtains length of $PQ$<br>Follow through from 'their' coordinates of $P$ and $Q$<br>(Does not need to be in the required form)                                                                                                                                                            | AO1.1b | A1F   |                                                                                                              |
|     | Sets out a well-constructed mathematical argument, using precise statements and correct use of symbols throughout to show the correct required result in required form<br><br>Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips | AO2.1  | R1    |                                                                                                              |
|     |                                                                                                                                                                                                                                                                                           |        |       | <b>Total 8 marks</b>                                                                                         |

**Q9.**

|     | Marking Instructions                                                                     | AO     | Marks | Typical Solution                                                                                               |
|-----|------------------------------------------------------------------------------------------|--------|-------|----------------------------------------------------------------------------------------------------------------|
| (a) | Uses either of the given coordinates in the given equation (accept product of the roots) | AO1.1a | M1    | $k = 4(2 + \sqrt{5}) - (2 + \sqrt{5})^2 = -$<br><br><b>ALT</b><br>$k = 4(2 - \sqrt{5}) - (2 - \sqrt{5})^2 = -$ |
|     | Obtains the correct value of $k$                                                         | AO1.1b | A1    | <b>ALT</b><br>$k = (2 - \sqrt{5})(2 + \sqrt{5}) = -1$                                                          |
| (b) | Sketches a graph with the correct shape ✓                                                | AO1.2  | B1    |                             |
|     | Deduces correct relative positioning of intersections with axes (must see labels)        | AO2.2a | B1    |                                                                                                                |
|     | Deduces minimum lies to right of y-axis in fourth quadrant                               | AO2.2a | B1    |                                                                                                                |
|     |                                                                                          |        |       | <b>Total 5 marks</b>                                                                                           |

**Q10.**

|     | Marking Instructions                                                                                              | AO     | Marks | Typical Solution                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------------|--------|-------|-------------------------------------------------------------------------------------|
| (a) | Finds a difference between 2 terms                                                                                | AO3.1a | M1    | $3e^p - 5 = 5 - 3e^{-p}$ (*)<br>$3e^p - 10 + 3e^{-p} = 0$                           |
|     | Forms an equation using two differences                                                                           | AO3.1a | M1    | $3e^{2p} - 10e^p + 3 = 0$                                                           |
|     | Forms a quadratic equation in $e^p$                                                                               | AO1.1a | M1    | $e^p = \frac{1}{3}, 3$                                                              |
|     | Obtains a correct quadratic equation                                                                              | AO1.1b | A1    | $p = \ln \frac{1}{3}, \ln 3$<br><b>ALT to (*)</b>                                   |
|     | Obtains 2 correct solutions for $e^p$ from 'their' quadratic<br>FT only applies if previous mark has been awarded | AO1.1b | A1F   | $2(5 - 3e^{-p}) = 3e^p - 3e^{-p}$<br><b>Or</b><br>$2(3e^{-p} - 5) = 3e^p - 3e^{-p}$ |
|     | Obtains final answers in an exact form<br>FT applies if previous mark                                             | AO2.2a | A1F   |                                                                                     |

|                       |                                                                                                  |        |    |                                                                                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------|--------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | has been awarded                                                                                 |        |    |                                                                                                                                                                                    |
| (b)                   | Finds a ratio between two consecutive terms<br>(no requirement to use $a$ and $r$ )              | AO3.1a | M1 | Assume it is possible that $3e^{-q}$ , 5 and $3e^q$ are three consecutive terms of a geometric sequence                                                                            |
|                       | Compares two ratios<br>(could be ratios of successive terms, no requirement to use $a$ and $r$ ) | AO3.1a | M1 | $a = 3e^{-q}$ , $ar = 5$ , $ar^2 = 3e^q$<br>$\frac{ar}{a} = \frac{5}{3e^{-q}} \Rightarrow r = \frac{5e^q}{3}$<br>$\frac{ar^2}{ar} = \frac{3e^q}{5} \Rightarrow r = \frac{3e^q}{5}$ |
|                       | Identifies a contradiction                                                                       | AO2.1  | R1 | $\frac{5}{3e^{-q}} = \frac{3e^q}{5} \Rightarrow 25 = 9$                                                                                                                            |
|                       | Draws a conclusion about $k$                                                                     | AO2.4  | R1 | This is a contradiction therefore $3e^{-q}$ , 5 and $3e^q$ cannot form three consecutive terms of a geometric sequence.                                                            |
| <b>Total 10 marks</b> |                                                                                                  |        |    |                                                                                                                                                                                    |

**Q11.**

|     | Marking Instructions                                                                                       | AO     | Marks | Typical Solution                                    |
|-----|------------------------------------------------------------------------------------------------------------|--------|-------|-----------------------------------------------------|
| (a) | Obtains correct horizontal component of the initial velocity                                               | AO1.1b | B1    | $2.5U = 40$<br>$U = 16$                             |
|     | Forms equation to find vertical component of initial velocity                                              | AO3.3  | M1    | $-10 = 2.5V - 0.5 \times 9.81 \times 2.5^2$         |
|     | Obtains correct vertical component of initial velocity                                                     | AO1.1b | A1    | $V = 8.2625$                                        |
|     | Forms equation for vertical component of velocity at height 3 using 'their' derived values for $U$ and $V$ | AO3.4  | M1    | $v_y^2 = 8.2625^2 + 2 \times (-9.81) \times 3$      |
|     | Obtains correct component of velocity                                                                      | AO1.1b | A1    | $v_y = 3.067...$                                    |
|     | Correct final speed with units, correct for 'their' $U$ and $v_y$<br>FT applies only if both M1            | AO3.2a | A1F   | $v = \sqrt{16^2 + 3.067^2} = 16.3 \text{ m s}^{-1}$ |

|                      |                                                                                                                                               |        |     |                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | marks have been awarded                                                                                                                       |        |     |                                                                                                                                                                                                                                             |
| (b)                  | States 'their' value of horizontal component of the initial velocity from part (a)                                                            | AO3.4  | A1F | 16 m s <sup>-1</sup>                                                                                                                                                                                                                        |
| (c)                  | Explains that horizontal velocity has been assumed to be constant in their model and that this is not likely to be true, with valid reasoning | AO3.5b | E1  | It was assumed that there were no resistance forces acting on the ball which is unlikely to be true in reality. The horizontal speed of the ball is likely to vary... air resistance would slow the ball down, wind might speed the ball up |
| <b>Total 8 marks</b> |                                                                                                                                               |        |     |                                                                                                                                                                                                                                             |

**Q12.**

(a) (i)  $(x - 3)^2$

*or  $p = 3$  seen*

$(x - 3)^2 + 2$

M1

A1

2

(ii)  $(x - 3)^2 = -2$

*FT their positive value of  $q$*

***not** use of discriminant*

M1

No (real) square root of  $-2$  therefore equation has no real solutions

*for graphical approach see below to see if SC1 can be awarded*

A1cso

2

(b) (i)  $x = \text{'their' } p \text{ or } y = \text{'their' } q$

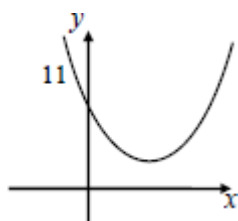
*or  $x = 3$  found using calculus*

M1

Vertex is at (3, 2)

A1cao

(ii)



*y* intercept = 11 **stated or marked on y-axis**  
(as *y* intercept of any graph)

*U* shape (generous)

B1

M1

*above x-axis* , vertex in first quadrant  
*crossing y-axis into second quadrant*

A1

3

(iii) Translation

*and no other transformation*

E1

through  $\begin{bmatrix} -3 \\ -2 \end{bmatrix}$

*FT negative of BOTH 'their' vertex coords*

M1

**both** components **correct** for A1; may  
describe in words or use a column vector

A1

3

[12]

**Q13.**

(a)  $x = 0 \Rightarrow y^2 - 4y - 12 = 0$

*sub  $x = 0$  & correct quadratic in  $y$*

*or  $(y - 2)^2 = 16$  or  $(y - 2)^2 - 16 = 0$*

M1

$(y - 6)(y + 2) (= 0)$

*correct factors or formula as far as*  $\frac{4 \pm \sqrt{64}}{2}$

$$\text{or } y - 2 = \pm \sqrt{16}$$

A1

$$\Rightarrow y = -2, 6$$

condone (0, -2) & (0, 6)

A1

3

(b)  $(x + 3)^2 - 9 + (y - 2)^2 - 4 (=12)$

**correct** sum of square terms **and** attempt to complete squares ( or multiply out) PI by next line

M1

$$(r^2 =) 9 + 4 + 12$$

$(r^2 =) 25$  seen on RHS

A1

$$(\Rightarrow r =) 5$$

$r = \sqrt{25}$  or  $r = \pm 5$  scores A0

A1

3

(c) (i)  $(CP^2 =) (2 - 3)^2 + (5 - 2)^2$

condone one sign slip within one bracket

M1

$$\Rightarrow (CP =) \sqrt{34}$$

$$n = 34$$

A1

2

(ii)  $PQ^2 = CP^2 - r^2 = 34 - 25$

Pythagoras used correctly with values FT "their"  $r$  and  $CP$

M1

$$(\Rightarrow PQ =) 3$$

A1

2

[10]

**Q14.**

(a)  $2x^2 - x - 1 = 2kx - 3k$   
 $2x^2 - x - 1 - 2kx + 3k = 0$  OE  
 $\Rightarrow 2x^2 - (2k + 1)x + 3k - 1 = 0$

*equated and multiplied out and all 5 terms on one side and = 0*

**AG** (correct with no trailing = signs etc)

B1

1

(b) (i)  $(2k + 1)^2 - 4 \times 2(3k - 1)$   
*clear attempt at  $b^2 - 4ac$*

M1

$(2k + 1)^2 - 4 \times 2(3k - 1) > 0$

*discriminant > 0 which must appear before the printed answer*

B1

$4k^2 + 4k + 1 - 24k + 8 > 0 \Rightarrow 4k^2 - 20k + 9 > 0$

**AG** (all working correct with no missing brackets etc)

A1cso

3

(ii)  $4k^2 - 20k + 9 = (2k - 9)(2k - 1)$   
*correct factors or correct use of formula as far as*  

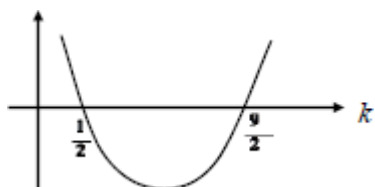
$$\frac{20 \pm \sqrt{256}}{8}$$

M1

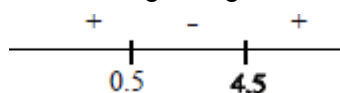
critical values are  $\frac{1}{2}$  and  $\frac{9}{2}$

*condone  $\frac{4}{8}$ ,  $\frac{36}{8}$  etc here but must combine sums of fractions*

A1



*sketch or sign diagram including values*



M1

$$k < \frac{1}{2}, \quad k > \frac{9}{2}$$

*fractions must be simplified condone use of OR but not AND*

A1

**Take their final line as their answer.**

4

[8]

**Q15.**

(a) (i)  $2(x + 1.5)^2$   
OE

M1

$$2(x + 1.5)^2 + 0.5$$

$$2\left(x + \frac{3}{2}\right)^2 + \frac{1}{2} \quad \text{OE}$$

A1

2

(ii) (Minimum value is) 0.5  
*ft their q*

B1✓

1

(b) (i)  $(AB^2 =) (x + 3)^2 + (3x + 9 - 5)^2$   
*condone one sign error inside one bracket*

M1

$$(3x + 4)^2 = 9x^2 + 24x + 16$$

OE

B1

$$AB^2 = x^2 + 6x + 9 + 9x^2 + 24x + 16 = 10x^2 + 30x + 25$$

$$\Rightarrow AB^2 = 5(2x^2 + 6x + 5)$$

**AG**

A1cso

(ii) *Either  $\sqrt{5 \times \text{'their' (a)(ii)}}$  or  $5 \times \text{'their' (a)(ii)}$   
using their minimum value from (a)(ii) and 5  
provided "their" (a)(ii) > 0*

M1

( Minimum length of  $AB = ) \quad \frac{1}{2}\sqrt{10}$

A1cso

2

[8]

**Q16.**

(a)  $(k - 2)^2 - 4 \times (2k - 7)(k - 3)$

*discriminant – condone one slip –  
condone omission of brackets*

M1

$$k^2 - 4k + 4 - 4(2k^2 - 6k - 7k + 21)$$

A1

“ their ”  $-7k^2 + 48k - 80 \geq 0$

*real roots condition;  $f(k) \geq 0$  must appear before final line*

B1

$$7k^2 - 48k + 80 \leq 0$$

**AG** (all working correct with no missing brackets etc)

A1cso

4

(b)  $7k^2 - 48k + 80 = (7k - 20)(k - 4)$

*correct factors*

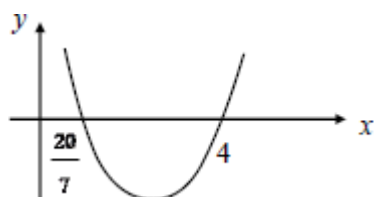
(or roots unsimplified)  $\frac{48 \pm \sqrt{64}}{14}$

M1

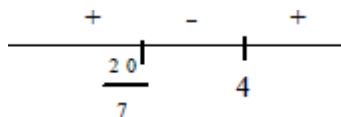
critical values are 4 and  $\frac{20}{7}$

*accept  $\frac{56}{14}$ ,  $\frac{40}{14}$  etc here*

A1



sketch or sign diagram including values



M1

$$\frac{20}{7} \leq k \leq 4$$

**Take their final line as their answer**

*fractions must be simplified here*

A1cao

4

[8]

**Q17.**

(a)  $b = a^c$

B1

1

(b)  $2 \log_2 (x + 7) - \log_2 (x + 5) = 3$

$$\log_2 (x + 7)^2 - \log_2 (x + 5) = 3$$

*A law of logs used correctly on a correct expression.*

M1

$$\log_2 \frac{(x+7)^2}{x+5} = 3$$

*A further correct use of law of logs on a correct expression.*

M1

$$= 3 \log_2 2 = \log_2 2^3$$

$$\Rightarrow \frac{(x+7)^2}{x+5} = 2^3$$

$$3 = 3 \log_2 2 \text{ or } 3 = \log_2 2^3 (= \log_2 8) \text{ seen}$$

*or*

$$\text{eg } \log f(x) = 3 \Rightarrow f(x) = 2^3 (= 8) \text{ OE}$$

B1

$$\Rightarrow (x + 7)^2 = 8(x + 5)$$

*Correct equation having eliminated logs and fractions*

A1

$$\Rightarrow x^2 + 14x + 49 = 8x + 40$$

$$\Rightarrow x^2 + 6x + 9 (= 0)$$

A1

Since  $6^2 - 4(1)(9) = 0$ , (there is only) one value of  $x$   
(which satisfies the given equation).

OE

CSO Need conclusion which is also correctly justified

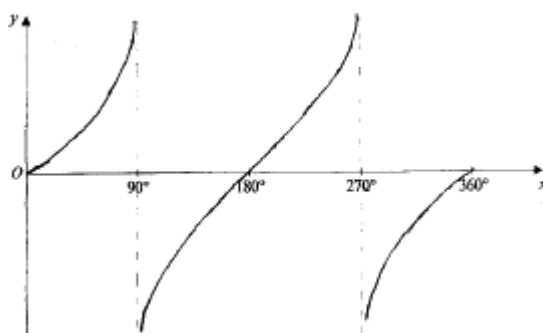
A1

6

[7]

**Q18.**

(a) (i)



*Ignore any part of the graph drawn outside interval  
 $0^\circ \leq x \leq 360^\circ$  in (a)*

*A 3 branch curve between 0 and 360 meeting the x-axis  
at or very close to 0, 180, 360 only*

B1

*A 3 branch curve between 0 and 360 with correct shape  
tending to infinity at, at least 3, of the 4 relevant ends*

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B1

*Correct graph for  $0^\circ \leq x \leq 360^\circ$  with correct intercepts.  
Asymptotes not explicitly required but graphs should  
show correct 'tendency' close to 90 and 270.*

B1

3

(ii)  $135^\circ$ ;  $315^\circ$

*B2 for both 135 and 315 and no 'extra' in interval  
 $0^\circ \leq x \leq 360^\circ$  (If not B2 then award B1 for either  
135 or 315 with or without extras)*

B2,1,0

2

(c) (i)  $6 \tan \theta \sin \theta = 5 \Rightarrow 6 \frac{\sin \theta}{\cos \theta} \sin \theta = 5$   
 $\tan \theta = \frac{\sin \theta}{\cos \theta}$  used

M1

$$6 \frac{\sin^2 \theta}{\cos \theta} = 5 \Rightarrow 6 \frac{1 - \cos^2 \theta}{\cos \theta} = 5$$

$\sin^2 \theta$  replaced by  $1 - \cos^2 \theta$  throughout

m1

$$6 - 6\cos^2 \theta = 5\cos \theta \Rightarrow 6\cos^2 \theta + 5\cos \theta - 6 = 0$$

Completion AG Be convinced

A1

3

(ii)  $6 \tan 3x \sin 3x = 5 \Rightarrow 6 \cos^2 3x + 5 \cos 3x - 6 = 0$   
 Using (b)(i) with  $\theta = 3x$  PI by attempting to solve  
 eg for theta then dividing soln(s) by 3

M1

$$(3\cos 3x - 2)(2\cos 3x + 3) (= 0)$$

$$(\cos 3x = 2/3, -3/2)$$

Correct factorisation or correct subst into the quadratic  
 formula PI by two 'correct' roots

m1

$$\cos 3x = \frac{2}{3} = \cos 48.1(89..) [= \cos \alpha]$$

$$3x = \alpha, 360 - \alpha, 360 + \alpha.$$

Dep on M1 only,

$3x = \alpha, 360 - \alpha, 360 + \alpha$  for c's  $\alpha$ . from an eqn  
 $\cos 3x = k$  where  $-1 < k < 1$  OE PI and no solns  
 from  $k$  outside  $-1 \leq k \leq 1$

m1

$$x = 16^\circ,$$

B1

$$104^\circ,$$

B1

136°

B1

AWRT 16, 104, 136. Deduct one mark (from any award of these 3 B marks) if more than three solns given inside the interval  $0^\circ \leq x \leq 180^\circ$ .

Ignore any solutions outside the interval  $0^\circ \leq x \leq 180^\circ$ .

NMS Max. is B3/6

6

[14]

**Q19.**

$$(8\sec x - 2\sec^2 x = \tan^2 x - 2)$$

$$8\sec x - 2\sec^2 x = \sec^2 x - 1 - 2$$

Using  $\tan^2 x = \sec^2 x - 1$  and NOT replacing  $\sec^2 x$  with  $1 + \tan^2 x$ .

M1

$$3\sec^2 x - 8\sec x - 3 (= 0)$$

A1

$$(3\sec x + 1)(\sec x - 3) (= 0)$$

Correct factors or correct use of quadratic equation formula or completing the square for 'their' equation.

$$\sec x - \frac{8}{6} = \pm \sqrt{\frac{64}{36} + 1}$$

m1

$$\text{Or } \sec x = \frac{8 \pm \sqrt{(-8)^2 - 4(3)(-3)}}{2(3)}$$

$$\sec x = 3, -\frac{1}{3} \text{ (or } -0.33)$$

Both correct.

A1

$$\sec x = \frac{1}{\cos x}$$

PI

B1

$$\left( \cos x = \frac{1}{3} \text{ or } 0.33 \right)$$

$$\left( \sec x = -\frac{1}{3} \text{ is impossible} \right)$$

$$x = 1.23, 5.05$$

*One correct. Must have earned A1 for correct quadratic, but independent of the second A1.*

A1

*Both correct and no extras in  $0 < x < 2\pi$ .  
CAO*

A1

[7]

**Q20.**

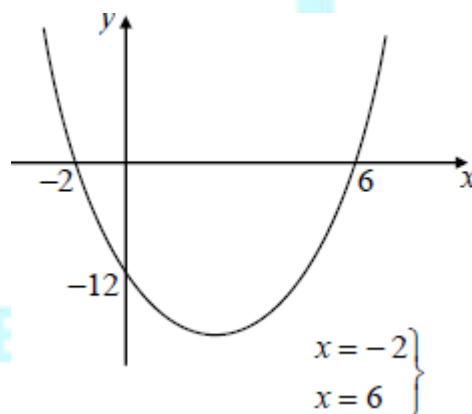
(a)  $(x - 6)(x + 2)$

*ISW for  $x = 6, x = -2$  etc*

B1

1

(b)



*correct  $x$  values or FT 'their' factors  
( $x$ -intercepts stated or marked on sketch) may  
be seen in (a)*

B1✓

$$y = -12$$

*(stated or  $-12$  marked on sketch)*

B1

*U - shaped curve  
approximately*

M1

“correct” shape in all 4 quadrants with minimum to right of y-axis

A1

(c) (i)  $(x - 2)^2$

$p = 2$

M1

$(x - 2)^2 - 16$

$p = 2$  and  $q = 16$

A1

(ii) (Minimum value is )  $-16$

*FT ‘their – q’*

B1✓

(d) Replacing each  $x$  by  $x + 3$   
**OR** adding 2 to their quadratic

*in original equation or ‘their’ completed square or  
 factorised form or replacing  $y$  by  $y - 2$*

M1

$y = [(x + 3)^2 - 4(x + 3) - 12] +$   
 $2$

or  $y = (x + 1)^2 - 14$

or  $y = x^2 + 2x - 13$

or  $y - 2 = (x - 3)(x -$   
 $5)$

**OE any correct equation in  $x$  and  $y$  *unsimplified***

A1

2

[10]

**Q21.**

(a)  $(x + 7)^2 + (y - 5)^2$

*one term correct ; condone  $(x - -7)^2$*

M1

*both terms correct with squares  
 and plus sign between terms*

A1

$$(x + 7)^2 + (y - 5)^2 = 5^2$$

*condone 25 for 5<sup>2</sup>*

A1cao

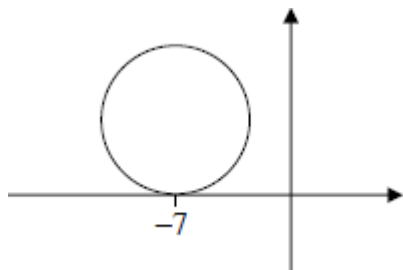
- (b) (i)  $C(-7, 5)$   
*correct or FT 'their' circle equation*

B1✓

- (ii)  $r = 5$   
*correct or FT 'their'  $r^2 > 0$*   
*condone  $\sqrt{25}$  etc but not  $\pm \sqrt{25}$*

B1✓

- (c) must draw axes



*freehand circle with C correct or FT 'their C' for quadrant of centre*

M1

*circle touching x-axis at -7 with -7 marked (need not show 5 on y-axis) but circle must not touch y-axis*

A1

- (d) (i)  $x^2 + (kx + 6)^2 + 14x - 10(kx + 6) + 49 = 0$   
*clear attempt to sub  $y = kx + 6$  into original or 'their' circle equation ...*

$$x^2 + k^2x^2 + 12kx + 36 + 14x - 10kx - 60 + 49 = 0$$

*...and attempt to multiply out*

M1

$$(1 + k^2)x^2 + 2kx + 14x + 25 = 0$$

$$\Rightarrow (k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$$

**AG** *condone  $x^2(1 + k^2) + 2x(7 + k) + \dots$  etc*

A1cso

- (ii) Equal roots ' $b^2 - 4ac = 0$ '

*allow statement alone if discriminant in terms of  $k$  attempted*

B1

$$[2(k + 7)]^2 - 4 \times 25(k^2 + 1)$$

*discriminant (condone one slip)*

M1

$$4\{k^2 + 14k + 49 - 25k^2 - 25\} = 0$$

$$-24k^2 + 14k + 24 = 0$$

$$\Rightarrow 12k^2 - 7k - 12 = 0$$

**AG** all working correct

*but = 0 must appear before last line*

A1

3

(iii)  $(4k + 3)(3k - 4)$

*correct factors or correct use of formula as far as*

$$k = \frac{7 \pm \sqrt{49 + 576}}{24}$$

M1

$$\Rightarrow k = -\frac{3}{4}, k = \frac{4}{3} \quad \text{OE}$$

are values of  $k$  for which line is a tangent

A1

2

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[14]

**Q22.**

(a) (i)  $\left(x - \frac{3}{2}\right)^2$

*or  $p = 1.5$  stated*

M1

$$\left(x - \frac{3}{2}\right)^2 + \frac{11}{4}$$

$$(x - 1.5)^2 + 2.75$$

A1

**Mark their final line as their answer**

2

(ii)  $x = \frac{3}{2}$

*correct or FT their “ $x = p$ ”*

B1 ✓

1

(b) (i)  $x^2 - 3x + 5 = x + 5 \Rightarrow x^2 = 4x$

*eliminating  $x$  or  $y$  and collecting like terms  
(condone **one** slip)*

*or  $(y - 5)^2 - 3(y - 5) + 5 = y \Rightarrow y^2 - 14y + 45 = 0$*

M1

$(x \neq 0) \Rightarrow x = 4$

A1

$y = 9$

A1

3

(ii)  $\frac{x^3}{3} - \frac{3x^2}{2} + 5x (+c)$

*one of these terms correct*

M1

*another term correct*

A1

*all correct (need not have  $+c$ )*

A1

3

(iii)  $\left[ \right]_0^4 = \frac{4^3}{3} - 3 \times \frac{4^2}{2} + 5 \times 4$

*must have earned M1 in part(b)(ii)*

*$F(\text{their } x_b) \{-F(0)\}$  “correctly sub’d”*

M1

$= 17\frac{1}{3}$

$\left( \frac{64}{3} - 24 + 20 = \right) \frac{52}{3} \text{ or } \frac{104}{6} \text{ etc}$

*condone 17.3 but not  $16\frac{4}{3}$  etc*

A1

Area trapezium  $\frac{1}{2}(x_B)(5 + y_B)$   
*FT their numerical values of  $x_B, y_B$*   
 $Area = \frac{1}{2} \times 4 \times 14 (= 28)$

B1 ✓

Area of shaded region  $28 - 17\frac{1}{3}$   
 $= 10\frac{2}{3}$   
*CSO;  $\frac{32}{3}$ , accept 10.7 or better*

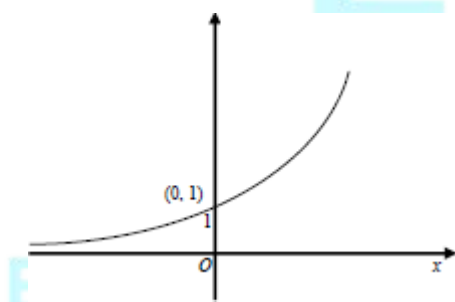
A1

4

[13]

**Q23.**

(a)



*Correct shape, curve in 1<sup>st</sup> two quadrants only, crossing positive y-axis once and asymptotic to negative x-axis.*

B1

*Coordinates (0, 1). Accept y-intercept indicated as 1 on diagram or stated as 'intercept = 1' B0 if graph clearly drawn crossing axes at more than one point*

B1

2

(b) (i)  $y^2 - 12 = y$  OE;  $7^{2x} - 12 = 7^x$  OE  
*Eliminates either  $x$  or  $y$  correctly*

M1

$(y - 4)(y + 3) (= 0); (7^x - 4)(7^x + 3) (= 0)$

Correct factors or  $y = \frac{1 \pm \sqrt{49}}{2}$  or better or  $7^x = \frac{1 \pm \sqrt{49}}{2}$   
or better

A1

Since  $y (= 7^x) > 0$ , [ $y (= 7^x) \neq -3$ ]  
(there is exactly one point of intersection)

*Clear indication that c's negative solution(s) has / have  
been considered and rejected*

E1

y-coordinate is 4

B1

4

(ii)  $7^x = 4$  so  $x \log 7 = \log 4$  [or  $x = \log_7 4$ ]

*OE ft on  $7^x = k$ , where  $k$  is positive, to either  
 $x \log 7 = \log k$  or  $x = \log_7 k$*

M1

$x = 0.712(414...) = 0.712$  to 3SF

*Condone > three significant figures. If use of logarithms not  
explicitly seen then score 0 / 2*

A1

2

[8]

Q24.

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(a)  $\frac{dy}{dx} = 18 + 6x - 12x^2$

*one of these terms correct*

M1

*another term correct*

A1

*all correct (no + c etc)  
(penalise + c once only in question)*

A1

3

(b)  $18 + 6x - 12x^2 = 0$

putting their  $\frac{dy}{dx} = 0$ ,

PI by attempt to solve or factorise

M1

$$6(3 - 2x)(x + 1) = 0$$

attempt at factors of **their quadratic**  
or use of quadratic equation formula

m1

$$x = -1, \quad x = \frac{3}{2} \quad \text{OE}$$

must see both values unless  $x = -1$   
is verified separately

A1

3

If M1 not scored, award SC B1 for verifying

that  $x = -1$  leads to  $\frac{dy}{dx} = 0$  and a further

SC B2 for finding  $x = \frac{3}{2}$  as other value

(c) (i)  $\frac{d^2y}{dx^2} = 6 - 24x$

FT their  $\frac{dy}{dx}$  but  $\frac{d^2y}{dx^2}$  must be correct if  
3 marks earned in part (a)

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B1 ✓

When  $x = -1$ ,  $\frac{d^2y}{dx^2} = 6 - (24 \times -1)$

Sub  $x = -1$  into their  $\frac{d^2y}{dx^2}$

M1

$$\frac{d^2y}{dx^2} = 30$$

A1cso

3

(ii) Minimum point

*must have a value in (c)(i)*

*FT "maximum" if their value of  $\frac{d^2y}{dx^2} < 0$*

E1✓

1

[10]

**Q25.**

(a) (i)  $y = \frac{1}{2}(7 - 3x)$

*attempt at  $y = \dots$*

*or use of 2 correct points using  $\frac{\Delta y}{\Delta x}$*

M1

$\Rightarrow \text{gradient} = -\frac{3}{2}$

*condone slip in rearranging if gradient is correct*

A1

2

(ii)  $y = \text{'their grad'} x + c$   
and substitution of  $x = 2, y = -7$

*or using  $3x + 2y = k$  with  $x = 2, y = -7$   
and attempt to find  $k$  or  $y - -7 = \text{'their grad'} (x - 2)$*

M1

$y = -\frac{3}{2}x + c, c = -4$

*correct equation in any form*

$y + 7 = -\frac{3}{2}(x - 2), 3x + 2y + 8 = 0, \text{etc}$

A1

$(x = 0 \Rightarrow) y = -4$

*or  $y$ -intercept = -4 or  $D(0, -4)$*

A1cso

3

(b)  $3x + 2(1 - 4x) = 7, y = 1 - \frac{4}{3}(7 - 2y)$

*elimination of  $y$  (or  $x$ ) (condone one slip)*

M1

$x = -1$

*one coordinate correct*

A1

$y = 5$

*other coordinate correct  
coordinates of  $A(-1, 5)$*

A1

3

(c)  $(5 - 2)^2 + (k + 7)^2 = 5^2$

*condone one sign slip within one bracket*

M1

(or  $k + 7 = 4$  or  $k + 7 = -4$ )

$k = -3$

*one correct value of  $k$*

A1

or  $k = -11$

*both correct (and no other values)*

A1

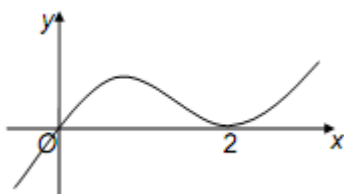
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[11]

**Q26.**

(a) (i)



*cubic curve with one max and one min  
(either way up)*

M1

*curve touching positive  $x$ -axis (either way up)*

A1

*correct graph passing through O and touching  
x-axis at 2*

A1

3

(ii)  $x(x^2 - 4x + 4) = 3$

$\Rightarrow x^3 - 4x^2 + 4x - 3 = 0$

*AG (must have = 0)*

B1

1

(b) (i)  $p(3) = 3^3 - 4 \times 3^2 + 4 \times 3 - 3$   
*p(3) attempted (condone one slip)*  
*NOT long division*

M1

$p(3) = 27 - 36 + 12 - 3$

$p(3) = 0 \Rightarrow -3$  is factor

*shown = 0 **plus statement***

A1

2

(ii) Either  $b = -1$  (coefficient of  $x$  correct)  
 or  $c = 1$  (constant term correct )  
*allow M1 for full attempt at long division or  
 comparing coefficients if neither  $b$  nor  $c$  is correct*

M1

$p(x) = (x - 3)(x^2 - x + 1)$

A1

2

(c) Discriminant of 'their quadratic' =  $(-1)^2 - 4$   
*numerical expression must be seen*

M1

Discriminant =  $-3$  (or  $< 0$ )  $\Rightarrow$  no real roots  
*must have correct quadratic and statement  
 and all working correct*

A1cso

(Only real root is  $x = 3$ )

B1

3

[11]

**Q27.**

(a) (i)  $(-)(x + 5)^2$

$q = 5$ ; condone  $(-x - 5)^2$

M1

$29 - (x + 5)^2$

$p = 29$  and  $q = 5$

A1

2

(ii)  $x = -5$  is line of symmetry

FT  $x = -$  'their  $q$ ' or correct

B1✓

1

(b) (i)  $4 - 10x - x^2 = k(4x - 13)$

$\Rightarrow x^2 + 4kx + 10x - 13k - 4 = 0$

Must see both these lines OE

$\Rightarrow x^2 + 2(2k + 5)x - (13k + 4) = 0$

AG all correct working and  $= 0$

B1

1

(ii) 2 distinct roots  $\Rightarrow b^2 - 4ac > 0$

stated or used (must be  $> 0$ )

B1

Discriminant  $= 4(2k + 5)^2 + 4(13k + 4)$

condone one slip (may be within formula)

M1

$4(4k^2 + 20k + 25 + 13k + 4) > 0$

or  $16k^2 + 132k + 116 > 0$

$$\Rightarrow 4k^2 + 33k + 29 > 0$$

*AG > 0 must appear before final line*

A1

3

(iii)  $(4k + 29)(k + 1)$

*correct factors or correct unsimplified quadratic*

*equation formula*  $\frac{-33 \pm \sqrt{33^2 - 4 \times 4 \times 29}}{8}$

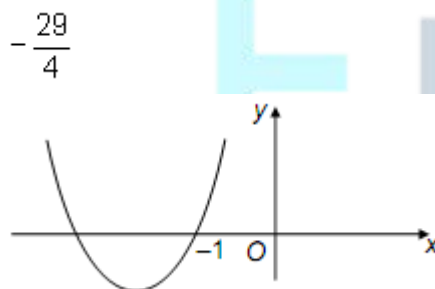
M1

$$k = -\frac{29}{4}, k = -1$$

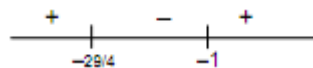
*condone  $k = -\frac{58}{8}$ , 7.25 etc but not left with*

*square roots etc as above*

A1



*sketch or sign diagram including values*



M1

$$k < -\frac{29}{4}, k > -1$$

*condone use of **OR** but not **AND***

A1

**Take their final line as their answer**

4

[11]

**Q28.**

(a) (i)  $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

*Condone numerator as  $\tan x + \tan x$*

B1

$$2 \tan x + \tan x (1 - \tan^2 x) = 0$$

*Multiplying throughout by their denominator*

M1

$$\tan x = 0$$

$$\text{or } (2 + 1 - \tan^2 x) = 0 \Rightarrow \tan^2 x = 3$$

**AG** Must show  $\tan x = 0$  and  $\tan^2 x = 3$

A1

3

**Alternative**

$$\tan 2x = \frac{\sin 2x}{\cos 2x} = \frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$$

$$\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x} + \frac{\sin x}{\cos x} = 0$$

(B1)

$$2 \sin x \cos^2 x + \sin x (\cos^2 x - \sin^2 x) = 0$$

$$\sin x (2 \cos^2 x + \cos^2 x - \sin^2 x) = 0$$

(M1)

$$\left. \begin{aligned} \Rightarrow \sin x = 0 \\ \Rightarrow \tan x = 0 \end{aligned} \right\} \text{ and } \left. \begin{aligned} 3 \cos^2 x = \sin^2 x \\ \tan^2 x = 3 \end{aligned} \right\}$$

(A1)

(3)

(ii)  $x = 60$  **AND**  $x = 120$

*Condone extra answers outside interval  
eg 0 and 180*

B1

1

(b) (i)  $2 \sin x \cos x = \cos x f(x)$

Where  $f(x) = \cos^2 x - \sin^2 x$   
or  $2\cos^2 x - 1$  or  $1 - 2\sin^2 x$

M1

$$2\sin x \cos x = \cos x(1 - 2\sin^2 x)$$

A1

$$(\cos x \neq 0) \quad 2\sin x = 1 - 2\sin^2 x$$

$$2\sin^2 x + 2\sin x - 1 = 0$$

**AG**

A1

3

$$(ii) \quad \sin x = \frac{-2 \pm \sqrt{4 - 4 \times 2 \times (-1)}}{2 \times 2}$$

*Correct use of quadratic formula or completing the square or correct factors*

M1

$$\sin x = \frac{-2 \pm 2\sqrt{3}}{4}$$

$\sqrt{12}$  must be simplified and must have  $\pm$ .

A1

$$\left. \begin{array}{l} \sin x = \frac{-1 - \sqrt{3}}{2} \text{ has no solution} \\ \sin x = \frac{\sqrt{3} - 1}{2} \end{array} \right\}$$

*Reject one solution and state correct solution.*

E1

3

[10]

**Q29.**

(a)  $(x + 2.5)^2$

$$p = \frac{5}{2}$$

B1

$$q = 7 - \text{'their' } p_2$$

*unsimplified attempt at  $q = 7 - \text{'their' } p_2$*

$$q = 7 - \frac{25}{4} = \frac{3}{4}$$

M1

$$(x + 2.5)_2 + 0.75$$

**mark their final line as their answer**

A1

3

(b) (i)  $x = -\text{'their' } p$  or  $y = \text{'their' } q$

or  $x = -\frac{5}{2}$  *cao found using calculus*

M1

$$\left(-\frac{5}{2}, \frac{3}{4}\right)$$

*condone correct coordinates stated*

$$x = -2.5, \quad y = 0.75$$

A1cao

2

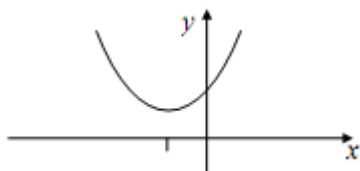
(ii)  $x = -\frac{5}{2}$

*correct or ft " $x = -\text{'their' } p$ "*

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1

(iii)



*y intercept = 7 stated*

*or seen in table as  $y = 7$  when  $x = 0$*

*or 7 marked as intercept on y-axis*

*(any graph)*

B1

*∪ shape*

M1

*vertex above x-axis in correct quadrant and  
parabola extending beyond y-axis into  
first quadrant*

A1

3

(c) Translation

*and no other transformation*

E1

through  $\begin{bmatrix} -\frac{5}{2} \\ 3\frac{3}{4} \end{bmatrix}$

*ft either 'their'  $-p$  or 'their'  $q$  or one component  
correct for M1*

M1

*both components correct for A1; may describe  
in words or use a vector*

A1cao

3

[12]

**Q30.**

(a)  $p(-1) = (-1)^3 - 2(-1)^2 + 3$

*$p(-1)$  attempted; not long division*

M1

$p(-1) = -1 - 2 + 3 = 0 \Rightarrow x + 1$  is a factor

*correctly shown = 0 plus statement*

A1cso

2

(b) (i) Quadratic factor  $(x^2 - 3x + 3)$

*$b = -3$  or  $c = 3$  by inspection*

*or full long division attempt*

*or comparing coefficients*

M1

$(p(x) =) (x + 1)(x^2 - 3x + 3)$

*must see correct product*

A1

2

(ii) Discriminant of quadratic

$$b^2 - 4ac = (-3)^2 - 4 \times 3$$

*'their' discriminant considered possibly within quadratic equation formula*

M1

$$\left. \begin{array}{l} b^2 - 4ac < 0 \Rightarrow \text{no real roots from quadratic} \\ \Rightarrow \text{only one real root} \end{array} \right\}$$

A1cso

2

[6]

**Q31.**

(a)  $(x - 3)^2 + (y + 8)^2$

*accept  $(y - -8)^2$*

B1

$= 100$

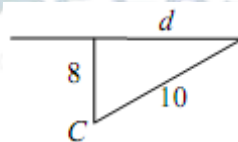
*condone  $RHS = 10^2$  or  $k = 10^2$*

B1

2

(b)  $y = 0 \Rightarrow \text{'their' } (x - a)^2 + b^2 = k$

**Alternative**



M1

$(x - 3)^2 = 36$  or  $x^2 - 6x - 27 (= 0)$  (PI)

A1

$\Rightarrow x = -3, 9$

$(d^2 =) 10^2 - 8^2$  M1

$d^2 = 36$  A1 or  $d = 6$

$\Rightarrow x = -3, 9$  A1

A1

- (c) Line  $CA$  has gradient  $-\frac{2}{5}$

M1

$CA$  has equation  $(y+8) = -\frac{2}{5}(x-3)$   
*any form of correct equation*  
 eg  $y = -\frac{2}{5}x + c, \quad c = -\frac{34}{5}$

A1

$$2x + 5y + 34 = 0$$

*integer coefficients – all terms on 1 side*

A1cso

3

- (d) (i) their  $(x-3)^2 + (2x+1+8)^2$  or  
 $x^2 + (2x+1)^2 - 6x + 16(2x+1) + 73$   
*substituting  $y = 2x + 1$  correctly into LHS of “their” circle equation and attempt to expand in terms of  $x$  only*

M1

$$x^2 - 6x + 9 + 4x^2 + 36x + 81 = 100$$

or  $x^2 + 4x^2 + 4x + 1 - 6x + 32x + 16 + 73 = 100$   
*any correct equation (with brackets expanded)*

A1

$$\Rightarrow 5x^2 + 30x - 10 = 0$$

*must see this line or equivalent*

$$\Rightarrow x^2 + 6x - 2 = 0$$

*AG; all algebra must be correct*

A1cso

3

- (ii)  $(x+3)^2 = 11$   
*or correct use of formula*

must get as far as  $x = \frac{-6 \pm \sqrt{44}}{2}$

M1

$x = -3 \pm \sqrt{11}$   
exactly this

A1cso

2

[13]

**Q32.**

(a)  $3 \ln x = 4$

$$\left( \ln x = \frac{4}{3} \right)$$

$$x = e^{\frac{4}{3}}$$

ISW. Condone  $\sqrt[3]{e^4}$

B1

1

(b)  $3 \ln x + \frac{20}{\ln x} = 19$

$$3(\ln x)^2 + 20 = 19 \ln x$$

correctly multiplying by  $\ln x$ .

M1

$$3(\ln x)^2 - 19 \ln x + 20 (= 0)$$

A1

$$(3 \ln x \pm 4)(\ln x \pm 5) (= 0)$$

use of formula, or completing the square  
must be correct

m1

$$\ln x = \frac{4}{3}, 5$$

A1

$$x = e^{\frac{4}{3}}, e^5$$

condone  $\sqrt[3]{e^4}$

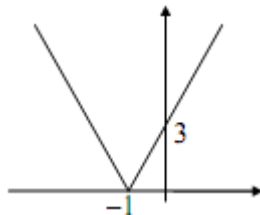
A1

5

[6]

**Q33.**

(a) (i)



*modulus graph, approximate V shape,  
touching negative x-axis and crossing y-axis*

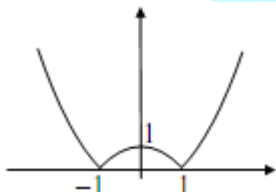
M1

*-1, 3 marked, graph symmetrical, straight lines*

A1

2

(ii)



*modulus graph in 3 sections, touching  
x-axis and crossing positive y-axis*

M1

$$\left. \begin{array}{l} \text{correct curvature} \\ \text{their } x > 1, \text{ their } x < -1 \\ \text{correct curve } -1 \leq x \leq 1 \\ \text{and } x = \pm 1, y = 1 \text{ marked} \end{array} \right\} \text{independent}$$

A1

A1

3

(b) (i)  $|3x + 3| = |x^2 - 1|$

$(3x + 3 = x^2 - 1)$

$(0 =) x^2 - 3x - 4$  —A  
*either A or B seen, all terms on one side*

M1

$x = 4, -1$

A1,A1

$(3x + 3 = 1 - x^2)$

$x^2 + 3x + 2 (= 0)$  —B

$x = -1, -2$

A1,A1

5

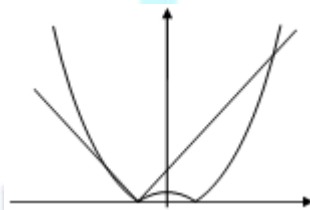
$\therefore x = -2, -1, 4$

SC NMS or partial method

$\left. \begin{array}{l} 1 \text{ correct value } 1 / 5 \\ 2 \text{ correct values } 2 / 5 \\ 3 \text{ correct values } 3 / 5 \end{array} \right\} \text{ independent of method mark}$

more than 3 distinct values max 2/5

(ii)



$x > 4, x < -2$

$x > \text{their largest}, x < \text{their smallest};$

CAO

M1,A1

2

[12]

**Q34.**

(a)  $x^2 - 8x + 15 + 2$

*Terms in x must be collected, PI*

B1

their  $x - 4^2$  (+ k)

*ft  $x - p^2$  for their quadratic*

M1

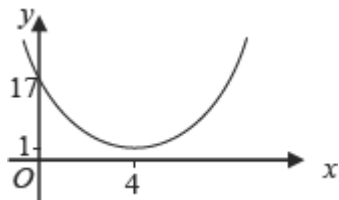
$$= x - 4^2 + 1$$

*ISW for stating if  $p = -4$  correct  
expression seen*

A1

3

(b) (i)



*U shape in any quadrant (generous)*

M1

*correct with min at (4, 1) stated or 4 and 1  
marked on axes  
condone within first quadrant only*

A1

*crosses y-axis at (0, 17) stated or 17  
marked on y-axis*

B1

3

(ii)  $y = k$

*$y = \text{constant}$*

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M1

$$y = 1$$

*Condone  $y = 0x + 1$*

A1

2

(c) Translation *(not shift, move etc)*  
*and no other transformation*

E1

with vector  $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$

*One component correct  
or ft either their  $p$  or  $q$*

M1

*CSO; condone 4 across, 1 up; or two separate vectors etc*

A1

3

[11]

**Q35.**

(a) (i)  $2(x - 5)^2$   
 $p = 5$

B1

+3

$q = 3$

B1

2

- (ii) Stating both  $(x - 5)^2 \geq 0$  and  $3 > 0$   
*FT their  $p$  &  $q$ , but must have  $q > 0$*

M1

$\Rightarrow 2x^2 - 20x + 53 > 0$  or  $2(x - 5)^2 + 3 > 0$

$\Rightarrow 2x^2 - 20x + 53 = 0$  has no real roots

*Must have statement and correct  $p$  &  $q$ .*

A1cso

2

(b) (i)  $b^2 - 4ac = (k + 1)^2 - 4k(2k - 1)$

*Condone one slip (including  $x$  is one slip)*

M1

$= -7k^2 + 6k + 1$

*Condone recovery from missing brackets*

A1

real roots  $\Rightarrow b^2 - 4ac \geq 0$

*Their discriminant  $\geq 0$  (in terms of  $k$ )*

$-7k^2 + 6k + 1 \geq 0$

*Need not be simplified & may earn earlier*

B1ft

$$\Rightarrow 7k^2 - 6k - 1 \leq 0$$

*AG (must see sign change)*

A1cso

4

(ii)  $(7k + 1)(k - 1)$

**Correct factors or correct use of formula**

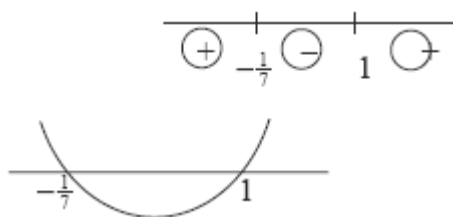
M1

Critical values  $k = 1, -\frac{1}{7}$

*May score M1, A1 for correct critical values seen as part of incorrect final answer with or without working.*

A1

Use of sign diagram or sketch



*If previous A1 earned, sign diagram or sketch must be correct for M1*

M1

*Otherwise M1 may be earned for an attempt at the sketch or sign diagram using **their** critical values.*

$$-\frac{1}{7} \leq k \leq 1$$

A1

Full marks for correct answer NMS

Condone  $-\frac{2}{14}$  throughout

Condone  $k \geq -\frac{1}{7}$  AND  $k \leq 1$  for full marks

**Take their final line as their answer.**

$$\left(-\frac{1}{7} < k < 1\right), \left(k \geq -\frac{1}{7} \text{ OR } k \leq 1\right),$$

$$\left(k \geq -\frac{1}{7}, k \leq 1\right) \text{ scores } M1A1M1A0$$

Answer only of  $k < -\frac{1}{7}, k < 1$  etc  
 scores M1, A1, M0 since the critical  
 values are evident.

Answer only of  $\frac{1}{7} \leq k \leq 1$  etc  
 scores M0, M0 since the critical values  
 are not both correct.



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